

Algebra 1
Unit 1
Lesson 7 Practice

Name _____
Date _____
Period _____

1. Write the expression $27^{\frac{2}{3}}$ in two different equivalent forms.

Expression 1	Expression 2

2. The value of $27^{\frac{2}{3}}$ is

- ☐ 9
- ☐ 32
- ☐ 54
- ☐ 72

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-125)^{\frac{2}{3}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

4. $-\left(\frac{3,125}{32}\right)^{\frac{2}{5}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

5. $(81)^{-\frac{3}{2}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

Evaluate each expression in questions 6-10 without a calculator.

6. $(4^{-2})(49^{-\frac{1}{2}})$

7. $(256^{\frac{1}{4}})^3 \cdot 6^{-1}$

8. $-\left(\frac{1}{4}\right)^{-\frac{7}{2}}$

9. $(-1^{15}) \cdot (-64^{\frac{1}{3}})$

10. $\left(\frac{5}{35}\right)^{-2} \cdot (2^{-2})$